

CYCLISTIC BIKE-SHARE ANALYSIS

Understanding Behavioral Differences Between Annual
Members and Casual Riders

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Business Problem

- Cyclistic aims to increase the number of annual memberships because long-term members generate more predictable revenue than casual riders.
- Before launching marketing campaigns, the company needs to understand how casual riders behave differently from annual members.

Objective

Analyze historical trip data to identify behavioral differences and recommend strategies to convert casual riders into annual members.

Dataset and Tools

Dataset

- Divvy (Cyclistic) Bike Share Dataset
- April 2020 – March 2021
- 3,478,810 rides
- 16 variables

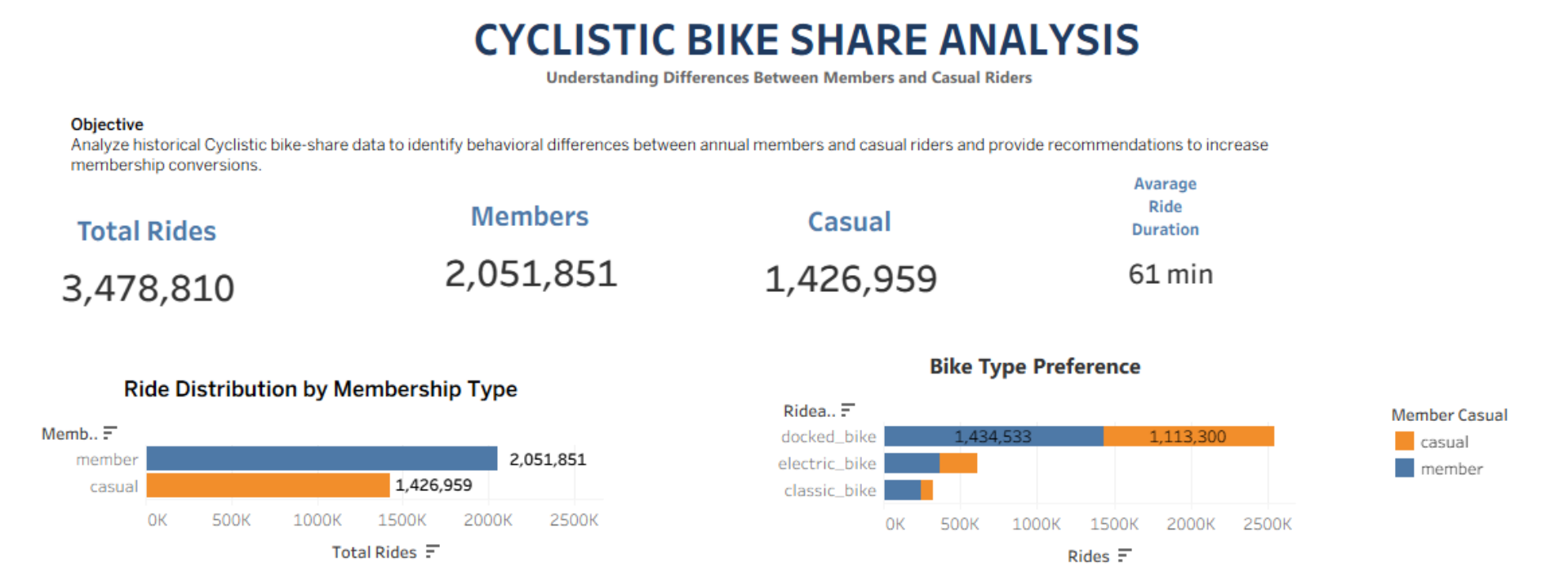
Tools

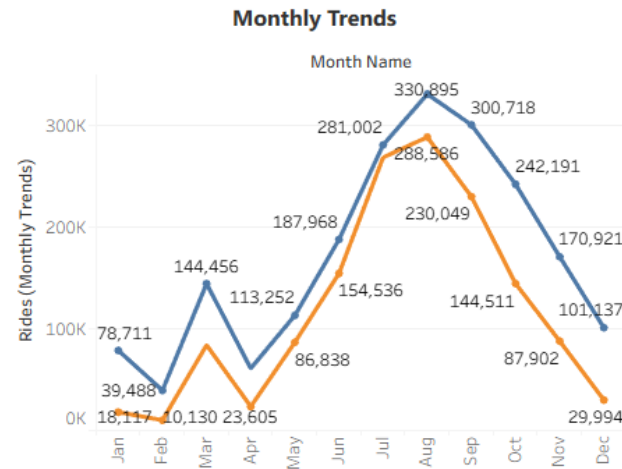
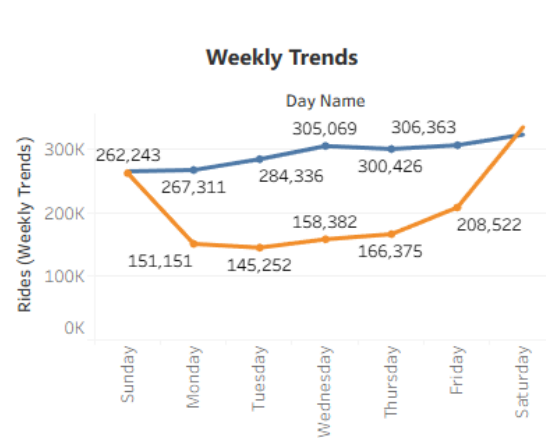
- PostgreSQL
- SQL
- pgAdmin 4
- Tableau
- GitHub

Data Preparation

- Imported twelve monthly CSV files into PostgreSQL.
- Verified all records after import.
- Checked for duplicate records.
- Investigated missing values.
- Created new analytical features:
 - Ride Length
 - Day of Week
 - Month
- Prepared clean data for visualization.

Dashboard Overview





KEY INSIGHTS

- Members generated 59% of all rides.
- Casual riders average nearly three times longer trips than members.
- Members primarily ride during weekdays, suggesting commuting behavior..
- Casual ridership peaks on weekends, especially Saturdays.
- Classic bikes are the most popular choice for both member types.

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The dashboard summarizes rider behavior using key performance indicators, bike preferences, weekly riding trends, monthly trends, and business insights.

Key Findings

❖ Finding 1

Annual Members completed 2,051,851 rides

❖ Finding 2

Casual Riders completed 1,426,959 rides

❖ Finding 3

Average Ride Duration

✓ Members → **16 minutes**

✓ Casual → **45 minutes**

❖ Finding 4

Casual riders are most active during weekends.

❖ Finding 5

Bike usage peaks during summer months.

Recommendations

- Launch weekend membership promotions.
- Introduce discounted annual membership packages during summer.
- Reward frequent casual riders with membership incentives.
- Promote financial savings associated with annual memberships.
- Target commuters through employer partnerships.

Expected Business Impact

- Increase annual membership subscriptions.
- Improve customer retention.
- Increase predictable long-term revenue.
- Support data-driven marketing decisions.
- Enhance customer engagement.

Conclusion

- The analysis demonstrates that annual members and casual riders exhibit distinct riding behaviors. Casual riders typically ride longer and more frequently during weekends, while annual members use the service more consistently throughout the week. These insights provide Cyclistic with actionable recommendations to improve membership conversion and support long-term business growth.

Thank You